

Koppesh P

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Summary

Computer Science undergraduate at VIT Chennai (CGPA 9.04) with hands-on experience designing and building backend services, REST APIs, and full-stack applications across internships and hackathons. Comfortable across Python, Java, and C++, with production experience in FastAPI/Django backends, MongoDB, and AWS-hosted deployments. Additional depth in applied AI/ML (RAG systems, vector search) used to extend traditional software with intelligent features.

Education

Vellore Institute of Technology, Chennai 2023 – 2027

B.Tech in Computer Science and Engineering (AI & ML) | CGPA: **9.04**

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks

VELS Vidyashram, Chennai 2022 – 2023

CBSE Class XII — **98.2%** | *School Topper*

Experience

AI & ML Intern – MHCognition Dec 2025 – Jan 2026

- **HRM Platform Backend Service:** Engineered and deployed a Django-based information-retrieval backend (TF-IDF search) and integrated it into an enterprise HRM platform, cutting repetitive HR support queries by ~20% and improving employee self-service access to documentation.
- **MH Cockpit Backend System:** Designed and built a production backend (FastAPI, MongoDB, Pinecone) exposing REST APIs for semantic document retrieval and LLM-based response generation (Gemini 2.5 Flash); deployed to AWS EC2/S3 and built an internal admin dashboard for lead tracking and usage analytics.

System Analyst Intern – LambdaDigital Pvt Ltd Jun 2024

- Performed requirement analysis and technology-stack evaluation for **tagminds.ai**, an Enterprise Asset Management SaaS platform, contributing to on-time Phase-1 delivery.

Projects

RoamIQ — Full-Stack AI Travel Planner VoyageTech Track Winner – NXTGEN Hackathon, SRM | 800+ applicants, 70+ teams

- Designed and built a full-stack web application (React.js frontend, Python backend, REST APIs) in a 24-hour hackathon sprint, shipping real-time weather adaptation, crowd analytics, accessibility scoring, and an emergency SOS feature.
- **Tech:** Python, React.js, REST APIs, LLM APIs, Prompt Engineering
- **Repository:** github.com/Kopichino/roamiq

UPACS — Unconscious Patient Autonomous Care System **5th Place** – VMeditthon Hackathon | 150+ teams

- Built an end-to-end monitoring application (Streamlit) integrating three ML models into a single real-time inference pipeline for mental-state detection, unconsciousness-level classification, and pain-score estimation.
- Implemented an ECG signal-processing and HRV feature-extraction pipeline with XGBoost-based inference, providing a low-cost alternative to EEG-based clinical monitoring.
- **Tech:** Python, TensorFlow, Scikit-learn, ECG Signal Processing, NumPy
- **Repository:** github.com/Kopichino/upacs

VoxCoder — Full-Stack Voice-Driven Web IDE

- Built a full-stack voice-driven web IDE (Next.js/React, Flask, **15+ REST endpoints**) with JWT auth, converting speech to code via OpenAI Whisper and Groq LLM.
- **Tech:** Next.js, React, TypeScript, Flask, REST APIs, SQLite, JWT Auth, OpenAI Whisper, Groq LLM
- **Repository:** github.com/Kopichino/VoxCoder

Skills

Programming Languages: Python, Java, C, C++, SQL

Backend & APIs: FastAPI, Django, Flask, REST API Design

Frontend: React.js

Cloud & DevOps: AWS (EC2, S3), Git, GitHub

Databases: MongoDB, Pinecone (Vector Database)

AI/ML: PyTorch, TensorFlow, Scikit-learn, XGBoost, LangChain, RAG, NLP, Prompt Engineering, Vector Search

Developer Tools: Streamlit, NumPy, Pandas

Certifications & Achievements

- **VoyageTech Track Winner** — NXTGEN Hackathon, SRM Institute of Science and Technology (800+ applicants, 70+ teams)
- **5th Place** — VMeditthon Hackathon, AI-based critical care system (150+ teams)
- **NPTEL** — Data Science for Engineers, Python for Data Science, Entrepreneurship (*Top 5% nationally, all three*)
- **FreeCodeCamp** — Responsive Web Design Certification